

Hospital Bed and ICU Management System

Introduction

This report explains the design of a hospital bed and ICU management system. The system helps hospital staff handle patients, check which beds are free, and give ICU places to the patients who need them most. The design is split into three layers, which are the presentation layer, the business layer and the data layer. Each layer has its own job.

Presentation Layer

The presentation layer is the part that the hospital staff can see and use. It has two components, the Hospital Staff Interface and the Bed and ICU Monitoring Interface. Staff use the first one to manage patients and ask for ICU places. The second one shows which beds and ICU rooms are free or full.

Business Layer

The business layer does the main work of the system. The Patient Triage Manager checks how serious each patient's condition is. The ICU Allocation Manager uses this to decide who gets an ICU place. The Bed and ICU Management Manager keeps track of all the beds and ICU rooms and updates their status.

Data Layer

The data layer stores all the information. It has one component, the Hospital Bed and Patient Database, which keeps the patient records and bed details. The three business layer components use the Data Access Interface to read and save data here.

How the Components Connect

The components talk to each other through interfaces. The Hospital Staff Interface uses the Patient Management Service, the ICU Allocation Service and the Bed and ICU Monitoring Service. The Triage Manager sends the Patient Priority Service to the ICU Allocation Manager, which then sends the Allocation Status Service to the Bed and ICU Management Manager. The monitoring screen gets its data from the Bed and ICU Management Manager.

Conclusion

In short, this design keeps the screens, the decision making and the data in separate layers. Because of this, a change in one layer does not break the others, and the system is easier to build, test and maintain.